

Deployment Planning — System Requirements & Scaling

1. Minimum Deployment (Pilot — 2.000 ONT)

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1 VM (All-in-One)

Semua service dalam 1 host:
 PostgreSQL + Redis + Dispatcher
 Dashboard + ACS Workers (fast, dl, ul)
 Direct Ping Worker (1 regional)

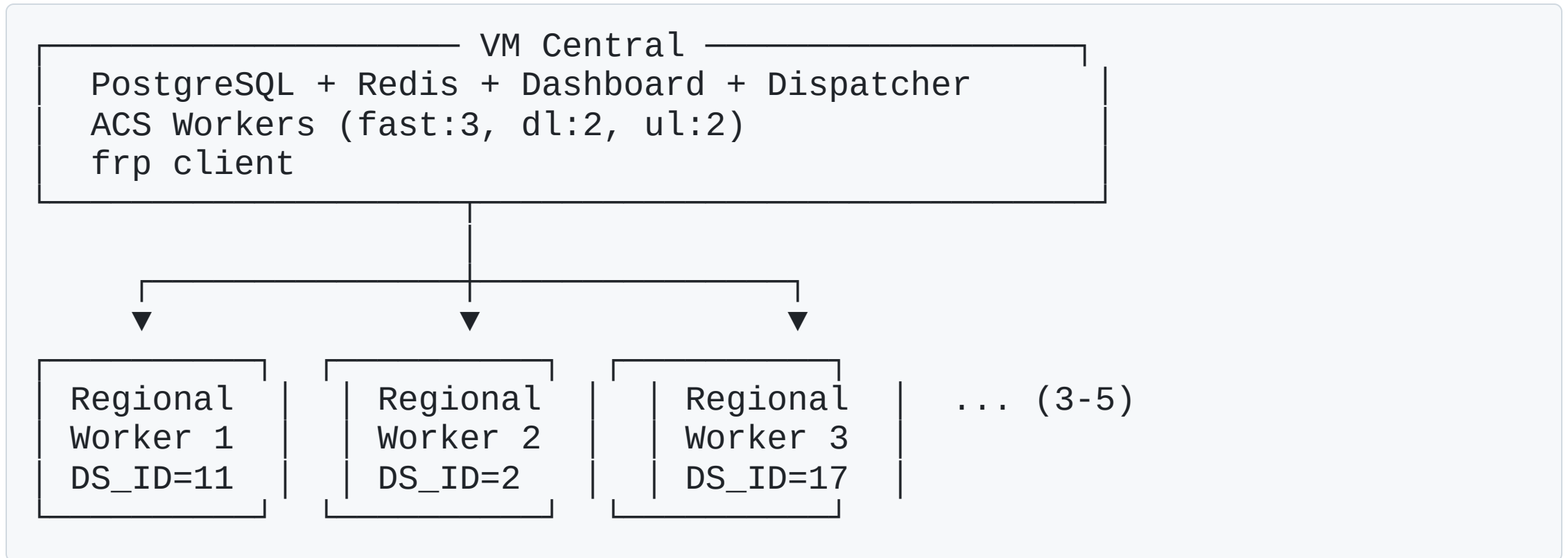
Akses: frp tunnel → gandooz.cloud:8804

VM Specification — Minimum

Item	Spec	Notes
CPU	4 vCPU	2.5 GHz+

2. Medium Scale (10.000 ONT)

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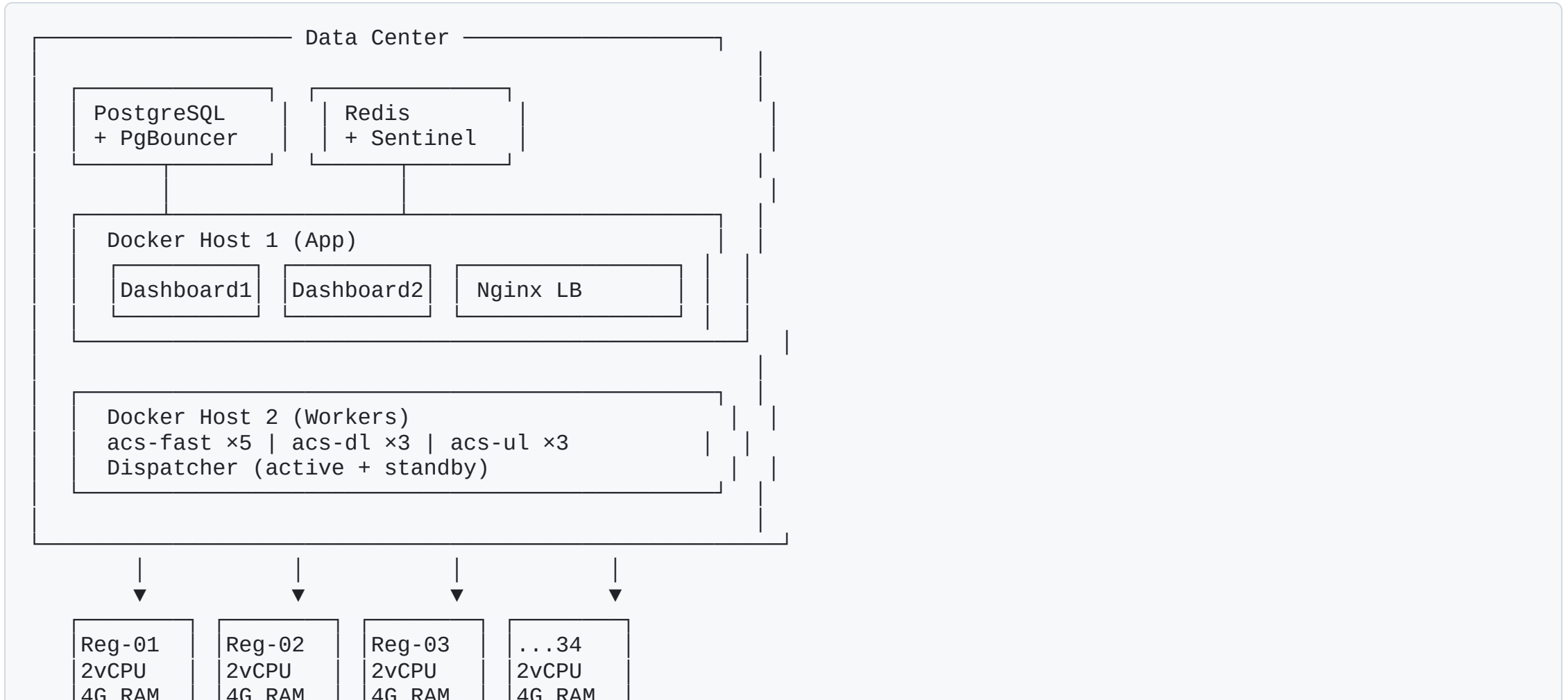


VM Specification

Cost of VM

3. Full Scale (26.000 ONT)

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4. Phased Scaling Plan

Phase 1: Pilot (0 → 2.000 ONT)

Timeline: Bulan 1

VM: 1 VM all-in-one + 1 regional worker

Budget:

1 VM (4vCPU/16GB/200GB) = ~\$80/bln

1 VM Regional (2vCPU/4GB) = ~\$30/bln

frp Server = ~\$10/bln

Total = ~\$120/bln

Test schedule:

- Ping: 4×/hari (setiap 6 jam)
- Download + Upload: 2×/hari (setiap 12 jam)

Trace computer: 1x/hari

5. Tabel Lengkap Per Phase

Parameter	Phase 1	Phase 2	Phase 3	Phase 4
ONT Devices	2.000	5.000	10.000	26.000
Timeline	Bulan 1	Bulan 2-3	Bulan 4-6	Bulan 7-12
Central VM	1	1	3	10
Regional VM	1	2	5	34 (12-15 host)
Total VM	2	3	8	44
Total vCPU	6	12	28	116
Total RAM	20 GB	40 GB	80 GB	230 GB
Total Storage	250 GB	600 GB	1.5 TB	4.3 TB
DB Size/30hr	~400 MB	~1 GB	~2 GB	~5 GB

6. Software Stack per VM

Central Servers

```
# Semua central server
OS: Ubuntu 22.04 LTS
Kernel: 6.2+ (HWE)
Docker: 24.0+ dengan Compose v2
Monitoring: prometheus-node-exporter (opsional)
Security: ufw, fail2ban, unattended-upgrades
Logging: journald + logrotate
Backup: pg_dump (cron harian) + WAL archiving
```

```
# DB Server tambahan
PostgreSQL 15 client tools
pgBackRest atau barman (jika butuh PITR)
```

```
# Regional Server
OS: Ubuntu 22.04 LTS minimal (no GUI)
Docker: 24.0+ (hanya perlu runtime)
SSH key-based auth (no password)
```

7. Cost Optimization

VM Sizing Guide

ONT Count	Central VM	Regional VM
2.000	4vCPU / 16GB	1× 2vCPU / 4GB
5.000	8vCPU / 32GB	2× 2vCPU / 4GB
10.000	3 host (total 20vCPU / 48GB)	5× 2vCPU / 4GB
26.000	10 host (total 48vCPU / 94GB)	12-15 host 34 worker containers

Tips Cost Saving

Strategy	Hemat	Cara
Spot/preemptible VM	50-70%	Untuk regional worker (stateless)
Reserved instance	30-40%	Untuk central (1 tahun)

8. Monitoring & Alerting Infra

Setiap VM harus di-monitoring:

```
# Install node_exporter untuk Prometheus
wget https://github.com/prometheus/node_exporter/releases/.../node_exporter.tar.gz
./node_exporter --web.listen-address=:9100 &

# Atau pakai telegraf (InfluxDB)
# Atau minimal: htop + nmon + dstat
```

Alert threshold per VM:

CPU > 80% selama 5 menit	→ WARNING
RAM > 90%	→ WARNING
Disk > 85%	→ WARNING → AUTO CLEANUP
Disk > 95%	→ CRITICAL
Docker container restart	→ WARNING
PostgreSQL connection > 80%	→ WARNING
Redis memory > 80%	→ WARNING

9. Database Migration Plan per Phase

Phase 1 → 2:

- Add indexes untuk performance
- Setup cron VACUUM ANALYZE

Phase 2 → 3:

- Setup PgBouncer connection pooling
- Tuning: shared_buffers, work_mem, effective_cache_size
- Partition test_results_* by month
- Archive data > 90 hari ke tabel_history

Phase 3 → 4:

- Setup PostgreSQL streaming replica
- WAL archiving untuk PITR
- pg_partman untuk auto-partitioning
- Consider read replica untuk dashboard queries

Phase 4 → Full:

- PgBouncer → multi-host connection routing
- Read/write splitting
- Query optimization untuk dashboard aggregate

10. Recommended Deployment Order

Bulan 1:

Week 1: Setup 1 VM → Docker → PostgreSQL + Redis

Week 2: Deploy Dashboard + frp → verify akses

Week 3: Deploy Dispatcher + Workers → test ping 100 ONT

Week 4: Deploy Direct Ping → test 500 ONT

Bulan 2:

Week 1-2: Scale ke 2.000 ONT → stabilkan

Week 3-4: Tambah regional worker ke-2 → 5.000 ONT

Bulan 3-4:

Split central ke multi-host

Setup PgBouncer + partition

Scale workers

Bulan 5-12:

Regional deployment bertahap

Tambah 3-5 province tiap bulan

Target: 26.000 ONT di bulan 12